

(1) Design a class Hierarchy (Pseudocode)

Class Person

    Name

    Age

    METHOD Get personDetails()

        INPUT Name

        INPUT Age

    END METHOD

END CLASS

CLASS Student INHERITS Person

    Register Number

    Marks[5]

    Average

    METHOD GetStudentDetails()

        CALL Get personDetails()

        INPUT Register Number.

        FOR i = 1 TO 5

            INPUT Marks[i]

        END FOR

    END METHOD

    METHOD CalculateAverage()

        Sum = 0

        FOR i = 1 TO 5

            Sum = Sum + Marks[i]

        END FOR

        Average = sum / 5

    END METHOD

END CLASS

```

CLASS    Research Student INHERITS Student
    Research Area
    Method GetResearchDetails()
        CALL GetStudentDetails()
        INPUT ResearchArea
    END METHOD
    METHOD DisplayCompleteDetails()
        PRINT Name
        PRINT Age
        PRINT RegisterNumber
        FOR I = 1 TO 5
            PRINT MARKS[I]
        END FOR
        PRINT Average
        PRINT Research Area
    END Method.
END class
MAIN
    CREATE RS. AS ResearchStudent
    CALL RS. GetResearchDetails()
    CALL RS. CalculateAverage()
    CALL RS. DisplayCompleteDetails()
END.

```

(2) Remove Duplicate Elements from an array

Problem statement:

Complete the missing pseudocode to remove duplicate elements from an integer array.

```

BEGIN
READ N
READ ARRAY A
FOR i ← 0 TO N-1
  FOR j ← i+1 TO N-1
    IF A[i] = A[j] THEN
      DELETE A[j]
      j ← j-1
      N ← N-1
    END IF
  END FOR
END FOR
PRINT "Modified Array"
FOR i ← 0 TO N-1
  PRINT A[i]
END FOR
END

```

Missing blank :

- (i)  $i \leftarrow 0$
- (ii)  $j \leftarrow i+1$
- (iii) THEN
- (iv) DELETE A[j]
- (v)  $N \leftarrow N-1$
- (vi)  $j \leftarrow j-1$

(3) Debug the pseudo code - Area of Square and Rectangle.

Task: Identify the logical errors and rewrite only the incorrect lines.

METHOD area (length)

RETURN length \* length

END METHOD

METHOD area (length, breadth)

RETURN length \* breadth

END METHOD

logical errors Identified!

- (1) Square area formula was incorrect
- (2) Rectangle method parameter should be length, breadth
- (3) Rectangle area formula should be length of breadth.
- (4) Trace the execution.

Problem statement: predict the output of the given pseudo code.

SUM  $\leftarrow$  0

FOR  $i \leftarrow$  1 to 5

IF  $i \text{ MOD } 2 == 0$  THEN

SUM  $\leftarrow$  sum +  $i$

ELSE

SUM  $\leftarrow$  sum \*  $i$

END IF

END FOR

PRINT SUM

END.

Trace table:

| i  | operation | sum |
|----|-----------|-----|
| 1. | 0 - 1     | -1  |
| 2. | -1 + 2    | 1   |
| 3  | 1 - 3     | -2  |
| 4  | -2 + 4    | 2   |
| 5  | 2 - 5     | -3  |

Output:

-3

(5) write efficient pseudocode - find the second largest distinct element.

BEGIN

READ N

FOR  $i \leftarrow 0$  TO  $N-1$

    READ  $A[i]$

END FOR

largest  $\leftarrow A[0]$

Second largest  $\leftarrow -1$

FOR  $i \leftarrow 1$  TO  $N-1$

IF  $A[i] > \text{largest}$  THEN

    Second largest  $\leftarrow \text{largest}$   
    largest  $\leftarrow A[i]$

ELSE IF  $A[i] > \text{second largest}$  AND

$A[i] \neq \text{largest}$  THEN

    Second largest  $\leftarrow A[i]$

END IF

END FOR

```
PRINT "second largest element = ",
```

```
SecondLargest
```

```
END.
```

6. Convert problem statement into pseudo code

```
BEGIN
```

```
INTERFACE Payment
```

```
    METHOD Pay (amount)
```

```
END INTERFACE
```

```
CLASS CreditCard IMPLEMENTS Payment
```

```
    METHOD Pay (amount)
```

```
        DISPLAY "Payment made using Credit  
card: ", amount
```

```
    END METHOD
```

```
END CLASS
```

```
CLASS UPI IMPLEMENTS Payment.
```

```
    METHOD pay (amount)
```

```
        DISPLAY "payment made using UPI: ",  
amount
```

```
    END METHOD
```

```
END CLASS
```

```
CLASS Net Banking IMPLEMENTS Payment
```

```
    METHOD pay (amount)
```

```
        DISPLAY "payment made using Net Banking:",  
amount
```

```
    END METHOD
```

```
END CLASS
```

```

READ choice
READ amount.
IF choice = 1 THEN
    Payment ← NEW CreditCard
ELSE IF choice = 2 THEN
    Payment ← NEW UPI
ELSE
    Payment ← NEW NetBanking
END IF
CALL Payment. Pay (amount)
END.

```

(7) Complete the pseudo code (Runtime Polymorphism)  
Problem :

Fill in the blanks to complete the inheritance  
and object creation.

```

CLASS Animal
    METHOD speak()
        PRINT "Animal"
    END METHOD
END CLASS

CLASS Dog INHERITS Animal
    METHOD speak()
        PRINT "Barks"
    END METHOD
END CLASS

Animal obj ← NEW Dog
obj.speak()

```

Output: Barks.

(8) Find Logical Errors (Constructors)

```
CLASS Employee
DATA
    id
    name
Constructor (id, name)
    this.id ← id
    this.name ← name
END CONSTRUCTOR
END CLASS
```

Logical Errors Corrected.

- (1) Constructor parameters id, name were missing
- (2) Assignment id - id was incorrect; corrected to this id ← id
- (3) Assignment name name was incorrect; correct to this . name ← name.

(9) Case study - Vehicle Management System.

```
BEGIN
CLASS Vechile
    Vechile NO
    Owner Name
    METHOD display()
        PRINT Vechile NO
        PRINT Owner Name
    ENDMETHOD
END CLASS
CLASS Car INHERITS Vehicle.
```

fuel type

METHOD display()

PRINT VehicleNo

PRINT OwnerName

PRINT fueltype

END METHOD

END CLASS

CLASS truck INHERITS vehicle

Load Capacity

METHOD display()

PRINT VehicleNo

PRINT VehicleNo

PRINT LoadCapacity

END METHOD

END CLASS

READ N

FOR I ← 1 TO N

READ type

IF type = "car" THEN

CREATE OBJECT V OF CAR

READ V. VehicleNo

READ V. OwnerName

READ V. fueltype

ELSE

CREATE OBJECT V OF truck

Read V. VehicleNo

Read V. OwnerName

READ V. LoadCapacity

(2) END IF

STORE V

END I FOR

FOR EACH V IN VehicleList

CALL V.display()

END FOR

END.